ADITYA SHIRALKAR

Computer Engineering Student

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ABOUT ME

Computer Engineering student with a strong interest in Software Development, Data Structures & Algorithms, and problem solving. Passionate about building practical projects and continuously improving coding skills through Java and LeetCode. Quick learner with good teamwork and leadership abilities, seeking internship and entry-level software engineering opportunities.

EDUCATION

Aug 24 - Ongoing

Savitribai Phule Pune University

B.Tech in Computer engineering | GPA : 8 | Third year

June 22 - June 24

Maharashtra State Board of Secondary and Higher Secondary Education (MSBSHSE)

HSC Boards : 83.33% | MHT-CET (PCM) : 93.33 percentile | JEE mains : 79.21 percentile

TECHNICAL SKILLS

Languages : Java, C++, Python, JavaScript, TypeScript.

Frameworks : Node.js/Express, HTML/CSS, Tailwind, Next.js, React.

Databases : MySQL, MongoDB.

Core Subjects: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems.

Tools & Technologies: Git, GitHub, VS Code, ESP8266, Arduino IDE, Docker, LLMs/RAG/agentic AI.

PROJECTS

Tesseract – Single-Pass 3D Model (SIH 2026)

- Converted a single-pass monocular drone video and GPS log into a georeferenced, textured 3D model within a 15-minute budget on one laptop GPU
- Built a full pipeline: frame screening, camera pose fitting (MapAnything + COLMAP), dense meshing and texturing with OpenMVS
- Implemented a gravity-constrained 5-DOF GPS fit in local ENU, cutting straight-pass scene error from 311 m to 0.041 m
- Delivered a three.js web workspace for viewing models and measuring distances, heights, and areas on the georeferenced surface
- Exports OBJ, PLY, LAS, GeoTIFF, GLB, FBX – each verified by read-back; built with a team of six

SovereignAgent (SwarajAI) – On-Premise AI Workbench

- Built an on-premise AI workbench that runs open-weight LLMs locally via Ollama, so sensitive data never leaves the network
- Implemented retrieval-augmented generation (RAG) over SOPs, inspection reports, and technical documents using vector search
- Added authentication with role-based permissions, audit logging, and egress monitoring to detect outbound requests
- Developed a model router for switching between LLMs and sandboxed tool execution for agent workflows
- Shipped with Docker Compose deployment and an Electron desktop wrapper

Smart Chef AI – Full-Stack Recipe Application

- Built a full-stack recipe app with a React/TypeScript frontend (Vite, TailwindCSS, shadcn/ui) and an Express backend
- Trained a recipe-matching model from CSV data to suggest recipes from available ingredients with filtering
- Designed interactive features: step-by-step cooking mode, chat interface, and camera-based ingredient scan page
- Added social features – user profiles, feeds, and achievements – backed by REST APIs

CustomX – Chrome Extension for Web Automation

- Developed a Manifest V3 Chrome extension (TypeScript, Vite) that records and replays browser sessions with timing preservation
- Built a custom selector engine for reliable element targeting across pages
- Implemented an authentication gate for managing login state and a rule-based page automation layer
- Wrote a test suite with Vitest and jsdom covering recorder, replayer, and selector modules

DMart Home Automation System (ESP8266)

- Developed an IoT home automation system on the ESP8266 for remote control of household appliances over Wi-Fi
- Focused the design on automation, ease of use, and energy efficiency
- Wrote firmware using Arduino IDE and C++ for wireless appliance switching
- Prepared full project documentation along with copyright-related submission